

Perception Challenge: Tread Depth

This take-home challenge is designed to assess your problem-solving approach and your knowledge of computer vision principles and techniques. Your task is to create a detailed design plan and implementation for a system that finds the tread depth of tires given a video clip of a camera panning across the tire.

As a deliverable we would like a document detailing your approach and some code showing off your approach. In order to save you time, a full and complete pipeline implementation is not required however we would like to see some code that demonstrates the techniques you would apply to the exercise. Use your own discretion on selecting what portion you choose to implement - our goal is for you to have fun and show off your skills to us.

We hope you will spend enough time on this project to demonstrate your skills, and we consider 2-4 hours nominal, but use your own judgement. Please try to get back to us within a week of receiving the challenge.

If you have any questions about the challenge or the process, please feel free to reach out.

What we're looking for

We are looking to understand:

- How you approach problems
- The breadth/depth of your knowledge, and your ability to apply it to a specific problem
- How you select a methodology and approach
- Thought process around iteration and trial and error
- Your ability to communicate with peers

You are welcome to submit any form of writing, and we aren't evaluating on how it's presented - feel free to do whatever you prefer! If it is helpful, you can respond directly in this document, share a GitHub repo or whatever else you'd prefer. Please be sure to email us when you are ready for us to take a look.

Please make sure to include the following information:

- What approach you would use to solve the problem, including details on specific methods and tools
 - Include why you have chosen to approach the problem the way you did
- Any additional information or data collection that you would request before actually implementing your approach
- How you would approach testing, validation and reliability of your solution
- Edge cases or subtle details that may come up and your approach for handling them
- Sensor selection, environmental constraints necessary, etc to achieve good results.
- Realistic estimates of implementation time/complexity/risk for your approach

Feel free to assume the audience is well-versed in computer vision.

Problem Description

For this challenge you should find a video file accompanying this document called *Tire Pan 2.mp4*. Given this file, please detail how you would develop a pipeline to measure the tread depth of tires in general, and implement a portion of the pipeline (or the whole thing, if you choose). If you have feedback about how you change the data collection process (change camera, lighting, motion, or alternative solutions), please detail that as well.